

Inter-VLAN Routing Basics Theory

Task 1 :- In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.

ANS :-

Create VLANs on Switch :

```
enable
configure terminal
vlan 10
name Foodies
vlan 20
name Shoppers
vlan 30
name Streamers
```

Assign Switch Ports to VLANs:

```
interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 20

interface fa0/3
switchport mode access
switchport access vlan 30
end
```

Connect PCs :

- PC-Zomato → Connect to Fa0/1 (Foodies VLAN)
- PC-Flipkart → Connect to Fa0/2 (Shoppers VLAN)
- PC-Spotify → Connect to Fa0/3 (Streamers VLAN)

Task 2 :- Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

ANS :-

Connect Switch to Router : Connect switch port Fa0/24 to router port Gi0/0/0.

Configure Trunk on Switch :

```
enable
configure terminal
interface fa0/24
switchport mode trunk
```

Configure Router Sub-interfaces :

```
enable
configure terminal
interface gi0/0/0
no shutdown

interface gi0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface gi0/0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

interface gi0/0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

Set PC Default Gateways:

- Foodies PC: IP 192.168.10.2 | Gateway 192.168.10.1
- Shoppers PC: IP 192.168.20.2 | Gateway 192.168.20.1
- Streamers PC: IP 192.168.30.2 | Gateway 192.168.30.1

Result : Ping between any PCs across different VLANs → Success!

Task 3 :- Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

ANS :-

1. Open PC Desktop : On each PC, go to Desktop → IP Configuration.

2. Set Settings for Each PC:

- Foodies PC (VLAN 10):
 - IP Address: 192.168.10.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
- Shoppers PC (VLAN 20):
 - IP Address: 192.168.20.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1
- Streamers PC (VLAN 30):
 - IP Address: 192.168.30.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.30.1

3. Verify : Open Command Prompt on Foodies PC and run ping 192.168.20.2
→ Success!

Task 4 :- Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity.

ANS :-

1. **Simulate Issue (Break it)** : On Router, delete or misconfigure sub-interface .30 :

```
enable
configure terminal
interface gi0/0/0.30
shutdown
```

Test: Ping from Shoppers PC (192.168.20.2) to Streamers PC (192.168.30.2) → **Fails** (Request timed out).

2. **Identify Problem** : Run show ip interface brief on Router. Interface gi0/0/0.30 is down/down or missing its IP address.
3. **Fix Misconfiguration** : Re-enable interface & verify settings :

```
interface gi0/0/0.30
no shutdown
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

4. **Verify Connectivity** : Ping from Shoppers PC (192.168.20.2) to Streamers PC (192.168.30.2) → **Success!**

Task 5 :- Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

ANS :-

Method	Advantages	Disadvantages
Router-on-a-Stick	1. Cheaper: Uses standard Layer 2 switches and 1 router. 2. Easier to secure: Router handles traffic control (ACLs) well	1. Slower: All traffic flows over one physical cable (bottleneck). 2. Single point of failure: If the router link breaks, all inter-VLAN traffic stops.
Layer 3 Switch	1. Super Fast: Routes traffic internally at hardware speed. 2. No Bottleneck: Doesn't rely on a single external router cable.	1. More Expensive: Layer 3 switches cost significantly more. 2. Less Advanced: Basic models have limited advanced routing and security features.